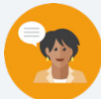


## 12.10 Solving One-Step Multiplication and Division Equations in One Variable Using Manipulatives

### Lesson Introduction

|                                                                                     |                     |                                                                                                                                                                                                                                                                                |                                                                                       |                                                                                                                                                                     |                                                                                  |                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Benchmark           | MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.                                                                                      |                                                                                       |                                                                                  | Learning Target                                                                  | <ul style="list-style-type: none"><li>I can solve one-step multiplication and division equations using manipulatives.</li></ul>                                                                                                                                     |
|    | Benchmark Coherence | Building On<br>MA.5.AR.2.4 Given a mathematical or real-world context, write an equation involving any of the four operations to determine the unknown whole number with the unknown in any position.                                                                          |                                                                                       | Working Toward<br>MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers. |                                                                                  |                                                                                                                                                                                                                                                                     |
|    | Essential Questions | <ul style="list-style-type: none"><li>How would you use algebra tiles to model and solve the multiplication and division equations?</li><li>How do you solve multiplication and division one-step equations?</li></ul>                                                         |                                                                                       |                                                                                                                                                                     |                                                                                  |                                                                                                                                                                                                                                                                     |
|    | Lesson Narrative    | In this lesson, students discover how to use algebra tiles to model and solve multiplication and division one-step equations. Students use inverse operations to isolate the variable while making sense of the meaning of the variable and the operation using algebra tiles. |                                                                                       |                                                                                                                                                                     |                                                                                  |                                                                                                                                                                                                                                                                     |
|  | Component Summary   | 12.10.1 Warm-Up (~5 min.)                                                                                                                                                                                                                                                      | Error analysis of image ( <i>SE p. 333</i> )                                          | 12.10.4 Guided Practice (10 min.)                                                                                                                                   | Practice modeling and solving one-step equations ( <i>SE p. 337</i> )            |                                                                                                                                                                                                                                                                     |
|                                                                                     |                     | 12.10.2 Exploration (10 min.)                                                                                                                                                                                                                                                  | Explore how to model and solve multiplication one-step equations ( <i>SE p. 333</i> ) | 12.10.5 Your Turn! (10min.)                                                                                                                                         | Additional practice modeling and solving one-step equations ( <i>SE p. 338</i> ) |                                                                                                                                                                                                                                                                     |
|                                                                                     |                     | 12.10.3 Guided Instruction (10 min.)                                                                                                                                                                                                                                           | Learn how to model and solve division one-step equations ( <i>SE p. 336</i> )         | 12.10.6 Wrap-Up (~5 min.)                                                                                                                                           | Write inverse operations sales pitch ( <i>SE p. 339</i> )                        |                                                                                                                                                                                                                                                                     |
|  | Resources           | Glossary                                                                                                                                                                                                                                                                       |                                                                                       | Materials                                                                                                                                                           |                                                                                  | Additional Preparation                                                                                                                                                                                                                                              |
|                                                                                     |                     | <u>Key Terms:</u> N/A<br><br><u>Additional Terms:</u> <ul style="list-style-type: none"><li>Inverse operations</li></ul>                                                                                                                                                       |                                                                                       | <ul style="list-style-type: none"><li>(Optional) Chart paper &amp; markers</li></ul>                                                                                |                                                                                  | <ul style="list-style-type: none"><li>Consider putting algebra tiles in zip top bags with the contents labeled for students to use or share.</li><li>If you do not have algebra tiles, you can create them using a template or use virtual algebra tiles.</li></ul> |

|                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <br>Teacher Tips              | <b>Common Misconceptions</b> <ul style="list-style-type: none"><li>Students may think that the coefficient represents the tens place and the variable represents the ones place (of a two-digit number).</li><li>Students may think the inverse operation of a negative coefficient is to add the coefficient to both sides (instead of dividing by it).</li><li>Students may not perform the inverse operation to both sides of the equation.</li><li>Students may struggle with the models for division equations.</li></ul> | <b>Things to Consider</b> <ul style="list-style-type: none"><li>Saying “2 multiplied by <math>x</math>” instead of “<math>2x</math>” will help students remember that the coefficient is multiplied by the variable.</li><li>If time is an issue consider assigning Your Turn! as homework.</li><li>The models for division equations may be difficult for students. Consider have strips of paper that students can divide up into parts. For example, for 12.10.3 question 1b provides the students with a strip of paper that is partitioned into 3 parts. Students can write <math>\frac{x}{3}</math> on each part and cut the strip. Then the student can use the paper with <math>\frac{x}{3}</math> to set equal to 4 and repeat that to demonstrate the model in question 1b.</li></ul> |
|                                                                                                                | <b>Literacy and Language Routines</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>Mathematical Thinking and Reasoning (MTR) Standard</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
|                                                                                                                | <b>Anticipate, Monitor, Select, Sequence, Connect</b> <p>Consider using this routine during 12.10.5 Your Turn! before class; think about the different ways students may represent and solve the problems. While students work, circulate and take notes on who used which strategy. Ask students who used different strategies to share their work. Sequence the strategies from least to most efficient. Facilitate a discussion to help students make connections among the strategies.</p>                                 | <b>MA.K12.MTR.2.1: Multiple Representations</b> <p>In 12.10.2 Exploration, students learn how to model and solve multiplication one-step equations using algebra tiles. In 12.10.3 Guided Instruction, students learn how to model and solve division one-step equations using algebra tiles. After learning how to represent and solve one-step equations using algebra tiles, students practice in the subsequent components.</p>                                                                                                                                                                                                                                                                                                                                                             |
| <br>Differentiate Instruction | Visual, auditory, and kinesthetic entry points are built in throughout this lesson. Using algebra tiles in 12.10.4 Guided Practice provides visual and kinesthetic entry points. Consider offering students the option to use virtual manipulative algebra tiles if they have access to tablets or computers. Utilizing algebra tiles to model and solve one-step algebraic equations builds conceptual understanding while encouraging a variety of learners to engage with the content in a meaningful way.                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Lesson Notes:                                                                                                  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |

## 12.10.1 Warm-Up: Math Fail

**Component Overview:** Students analyze an image to determine the “math fail.”



### 12.10.1 Warm-Up: Math Fail

This children’s book has published a “math fail.”

1. Explain the error shown in this image.



*There are 6 bananas in the image, but the writing shows there should be five.*



*Consider an extension for homework in which students have to find “math fails” in their world—at the grocery store, on a sale sign, or in a book.*

## 12.10.2 Exploration: Exploring Relationships

**Component Overview:** Students explore how to model and solve multiplication one-step equations using algebra tiles. Students develop a conceptual understanding of equations as balanced scales. This helps students understand why they need to perform inverse operations to both sides of the equation. Finally, students review the inverse operations for addition, subtraction, multiplication, and division.

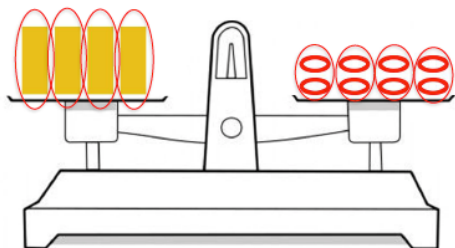


### 12.10.2 Exploration: Exploring Relationships

With your partner, complete the following.

1. Nadia made 8 friendship bracelets for her 4 best friends. Nadia wants to give all 8 of the bracelets away equally to each friend.
  - a. This situation can be modeled by the equation  $4 \times \square = 8$ . What operation is occurring in the equation?  
*Multiplication*
  - b. Using mental math, how many bracelets will each friend receive?  
*Each friend should receive 2 bracelets.*

This situation can also be represented using a balance scale. The rectangles represent Nadia's friends and the ovals represent the bracelets.



- c. Using the rectangles and ovals, sketch an image that demonstrates how many bracelets each friend will receive.  
*Answers will vary. The above diagram is a sample example.*
- d. What operation does your sketch demonstrate?  
*Division*



Auditory, visual, and kinesthetic entry points are provided throughout this 12.10.2 Exploration. The balance scale provides a visual to help students conceptually understand balancing equations. Algebra tiles can be used to model equations kinesthetically. Collaboration and discussion allow for auditory entry points as students verbally process each question.



As you circulate listen for misconceptions and incorrect vocabulary. The context is not difficult for students so that the students can focus on the mathematics they use and how that mathematics can be represented algebraically. Students who think they understand may want to speed through the questions without thinking. Keep an eye on the students who quickly go through question 1, but then struggle with question 2.

## 12.10.2 Exploration: Exploring Relationships (continued)

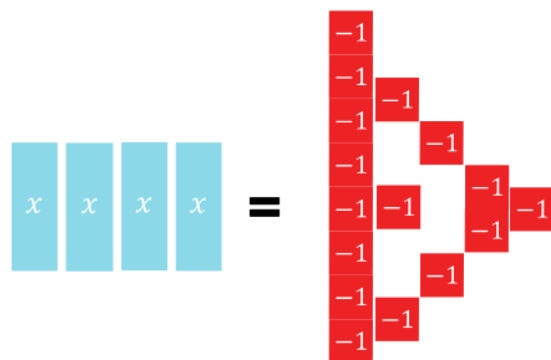
- e. How many bracelets will each friend receive?  
*2 bracelets*
- f. Discuss what you notice about the operation included in the equation and the operation you used to find the number of bracelets each friend will receive. Summarize your discussion.  
*Answers will vary. Multiplication and division are inverse operations.*

Recall that when solving an equation, the mathematical sentence must remain balanced so that the two expressions remain equivalent.

2. A model of the equation  $4x = -16$  is shown using algebra tiles.



The  tile represents the variable,  $x$ , and the  represents the constant term where each tile is 1 unit.



- a. What operation is being applied to the 4 and the  $x$  in the equation  $4x = -16$ ?  
*Multiplication*

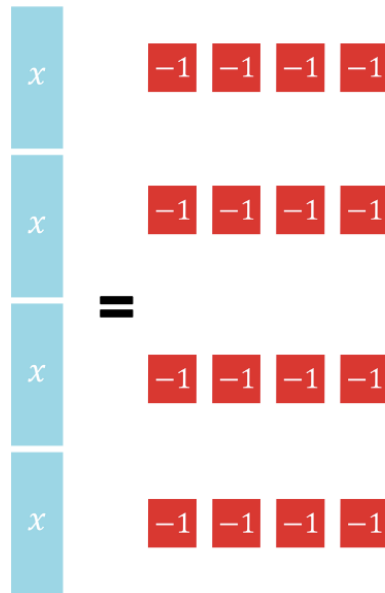


*Ask students how they think the variable can be isolated. This could be rephrased as grouping each side to show how many each  $x$  represents. Have students share their ideas and then test them using the algebra tiles. Students may be able to figure out how to solve the equation independently if given time to experiment with the algebra tiles.*

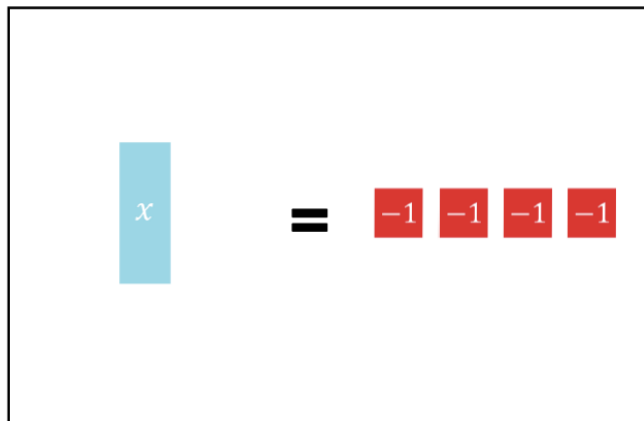
## 12.10.2 Exploration: Exploring Relationships (continued)

- b. The algebra tiles are arranged on the equation mat into equal-sized groups on both sides of the equation. How many groups were created?

*4 groups*



- c. Use algebra tiles to sketch the solution of the equation.



*Consider having students write the equation next to each step they modeled with the algebra tiles. Writing the equation next to the algebra tiles representation will help students make connections between the representations.*

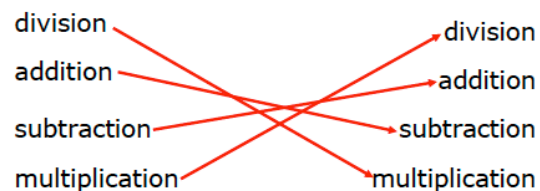
- d. What is the value of  $x$ ?

*$x = -4$*

## 12.10.2 Exploration: Exploring Relationships (continued)

3. Inverse operations can be helpful in solving equations.

- a. Draw an arrow to match each operation on the left with its inverse operation on the right.



- b. Explain what it means for operations to be inverses.

*Answers will vary. Inverse operations "undo" each other.*

- c. Which inverse operations were used to solve the equations in this exploration?

*Multiplication and division*



*If students are struggling to match, explain inverse operations by further scaffolding question 3 through questioning:*

- *What does "inverse" mean?*
- *How can you determine the inverse of an operation?*

## 12.10.3 Guided Instruction: Modeling the Inverse of Division

**Component Overview:** Students model and solve division one-step equations using algebra tiles. Students solve the equation by multiplying both sides of the equation by the number the variable was divided by. Remind students that they can check their solution by substituting it in for the variable and seeing if the left side equals the right side.



### 12.10.3 Guided Instruction: Modeling the Inverse of Division

Algebra tiles can also be used to model division equations.

1. The equation  $\frac{x}{3} = 4$  can be rewritten as  $x \div 3 = 4$ .

To represent division, show that  $x$  tile is being divided into equal parts. Sketch lines on the  $x$  tile to represent the fraction  $x \div 3$ .

This means when  $x$  is divided into 3 parts, each part is equal to 4. The model shown illustrates that  $\frac{x}{3} = 4$ .

$$\frac{x}{3} = 4$$

- a. If each  $\frac{1}{3}$  of  $x$  is equal to 4, discuss with your partner what the value of  $x$  is. As part of your discussion, include which operation can be used to determine the value of  $x$ . Summarize your discussion.

*Answers will vary.  $x$  is equal to 12. The problem is using division so the inverse that would need to be used is multiplication.*



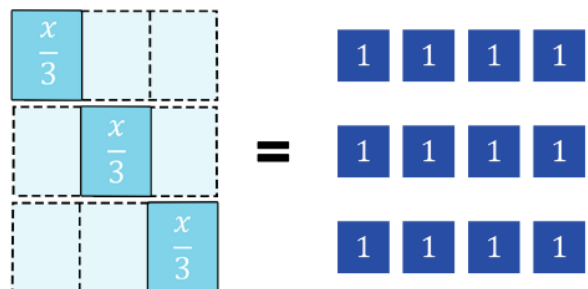
Consider creating an anchor chart with instructions, examples, and visual models for students to reference in subsequent components.



Students could say addition or multiplication. If students say addition, it would be repeated addition, which can be related to multiplication.

## 12.10.3 Guided Instruction: Modeling the Inverse of Division (continued)

- b. The image models the relationship discussed that can be used to determine the value of  $x$ .



Explain how this representation can be used to determine the value of  $x$ .

*Answers will vary.*

*One-third of the boxes is four.*

- c. What is the value of  $x$ ?  
 $x = 12$
- d. Complete the statements by writing operations on each blank.

The original equation involved the operation Division. Its inverse operation, Multiplication, was used to solve the equation.



Take the time to explain the model in question 1b. Point out how each  $\frac{x}{3} = 4$ . Then, help the student to see that the left side is three  $\frac{x}{3}$ 's which is equivalent to  $x$ , and since the left side has three  $\frac{x}{3}$ 's, the right side needs three sets of the four one-unit tiles. Lead the students so that they relate the model to multiplication.



To provide additional scaffolding, ask guiding questions like:

- What is the inverse operation of division?
- What is the inverse operation of multiplication?
- Why do you perform the inverse operations on both sides of the equation?
- How can you check your equation to make sure the solution is correct?

## 12.10.4 Guided Practice: Solving Equations Using Manipulatives

**Component Overview:** Students practice modeling and solving one-step equations using algebra tiles. Students can complete the problems independently or work with a partner. Facilitate a whole-class discussion at the end of the component to consolidate student understanding.



### 12.10.4 Guided Practice: Solving Equations Using Manipulatives

- Complete the table using algebra tiles to sketch the solution of  $-15 = 3x$  and explain each step. The first row has been completed.

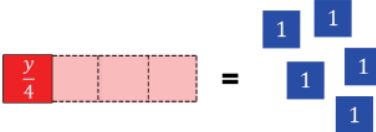
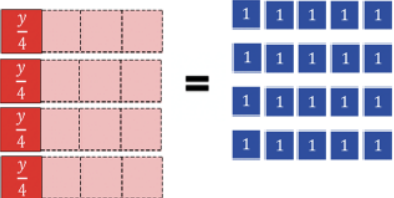
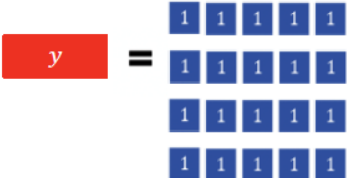
| Step   | Algebra Tiles | Description                                              |
|--------|---------------|----------------------------------------------------------|
| Step 1 |               | Use algebra tiles to represent the equation $-15 = 3x$ . |
| Step 2 |               | 3 groups of five $-1$ tiles = 3 groups of $x$ tiles      |
| Step 3 |               | $x = -5$                                                 |



Using algebra tiles provides visual and kinesthetic entry points. Consider offering students the option to use virtual manipulative algebra tiles if you they access to tablets or computers.

## 12.10.4 Guided Practice: Solving Equations Using Manipulatives (continued)

2. Complete the table using algebra tiles to sketch the solution of  $\frac{y}{4} = 5$  and explain each step. The first row has been completed.

| Steps  | Algebra Tiles                                                                      | Description                                                     |
|--------|------------------------------------------------------------------------------------|-----------------------------------------------------------------|
| Step 1 |   | Use algebra tiles to represent the equation $\frac{y}{4} = 5$ . |
| Step 2 |   | Four groups of $\frac{y}{4}$ is equal to four groups of 5.      |
| Step 3 |  | $y = 20$                                                        |



Step 2 is slightly different than question 1b in 12.10.3 Guided Instruction. In question 1b, the  $\frac{x}{3}$  was staggered to show the whole. Here the  $\frac{y}{4}$  is not staggered. Ensure the students understand that four  $\frac{y}{4}$ 's will result in  $1y$ . If necessary, explain the model to the students.

## 12.10.5 Your Turn!

**Component Overview:** Students practice modeling and solving one-step equations. If students feel confident, they can attempt to solve the equations without the algebra tiles. Ask students to check their work to ensure their solutions are correct.



### 12.10.5 Your Turn!

1. The equation  $10 = 5x$  is modeled on the equation mat.

$$\begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 1 \\ \hline 1 & 1 \\ \hline 1 & 1 \\ \hline 1 & 1 \\ \hline \end{array} = \begin{array}{|c|c|c|c|c|} \hline x & x & x & x & x \\ \hline \end{array}$$

Miranda told her classmate the value of  $x$  is  $\frac{1}{2}$  because  $5 \div 10 = \frac{1}{2}$ .

- a. Explain whether you agree with Miranda.  
*Answers will vary. Miranda is incorrect because she divided 5 by 10 when she should have divided 10 by 5 for an answer of 2.*
- b. Sketch the next step in determining the solution to the equation.

$$\begin{array}{|c|c|} \hline 1 & 1 \\ \hline 1 & 1 \\ \hline 1 & 1 \\ \hline 1 & 1 \\ \hline 1 & 1 \\ \hline \end{array} = \begin{array}{|c|} \hline x \\ \hline x \\ \hline x \\ \hline x \\ \hline x \\ \hline \end{array}$$



*Anticipate, Monitor, Select, Sequence, Connect*  
 Consider using this routine during this section. Before class, think about the different ways students may represent and solve the problems. While students work, circulate and take notes on who used which strategy. Ask students who used different strategies to share their work. Sequence the strategies from least to most efficient. Facilitate a discussion to help students make connections among the strategies.

## 12.10.5 Your Turn! (continued)

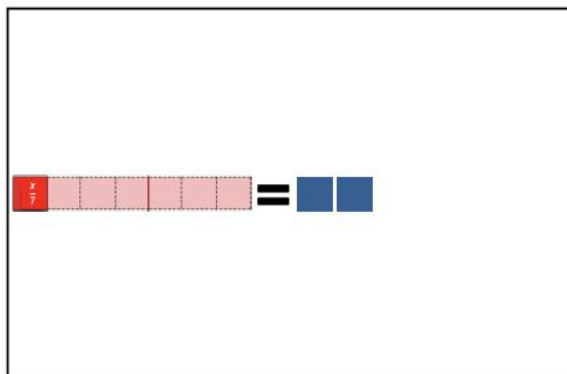
c. What is the value of  $x$  according to your sketch?

$$x = 2$$

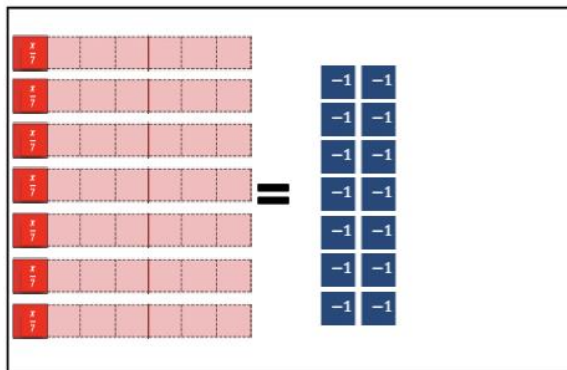
2. Consider the equation  $\frac{x}{7} = -2$ .

a. Using algebra tiles, sketch each step to find the solution of  $\frac{x}{7} = -2$ .

Step 1



Step 2



Consider providing algebra tiles to provide an additional kinesthetic entry point for students.

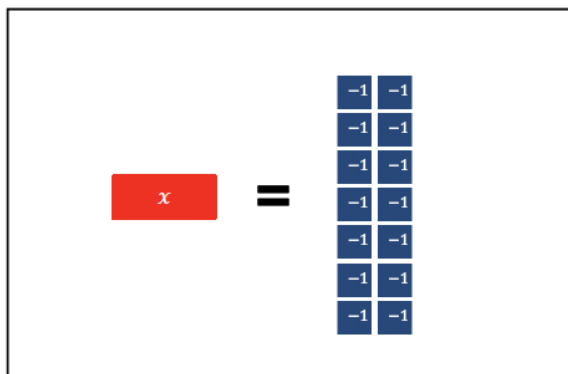


Students may struggle to complete question 2a, Steps 1-3 on their own. For students who need additional support, further scaffold through questioning such as:

- How can you represent  $\frac{x}{7}$ ?
- What is the inverse operation that must be performed to solve this equation?
- How do you represent that operation using algebra tiles?

## 12.10.5 Your Turn! (continued)

Step 3


$$x = \begin{array}{|c|c|} \hline -1 & -1 \\ \hline -1 & -1 \\ \hline -1 & -1 \\ \hline -1 & -1 \\ \hline -1 & -1 \\ \hline -1 & -1 \\ \hline -1 & -1 \\ \hline \end{array}$$

- b. What is the value of  $x$  according to your sketch?

$x = -14$



*If students finish early, ask them to create a real-world problem that could be modeled by the equations in this component. Another option is to have students write additional one-step equations, swap with a partner, and solve each other's equations.*

## 12.10.6 Wrap-Up: Cool Down

**Component Overview:** Students write an inverse operations sales pitch. Consider sharing an example to help students understand the expectations for this problem. Use data from this formative assessment to determine which students need remediation before the next lesson.



### 12.10.6 Wrap-Up: Cool Down

1. Create a sales pitch to explain inverse operations.  
*Answers will vary.*



*This activity provides an auditory entry point. Consider providing students with sentence stems to help them create their sales pitch or make it a competition, with the best sales pitch winning a prize.*

2. Share your sales pitch with a partner in 60 seconds or less.

*Answers will vary.*